



Passenger Flow Simulation

- BSCS – 3rd Year, 6th Semester

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The Analysis and AI Simulation of Passenger Flows in an Airport Terminal: A Decision-Making Tool

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Crowd Management

Risk Assessment

Simulation Model

ABSTRACT

A decision-making tool is proposed to handle passenger flows in airport terminals, built around a simulation model that can be updated and modified to match a terminal's current requirements. It is meant to help airport managers proactively mitigate risk and evaluate crowd-management strategies — eliminating overcrowding hazards and minimizing waiting times so travel stays seamless, safe and satisfying.

Overcrowding makes it hard for passengers to move freely and raises the risk of accidents, security problems and service interruptions — especially in an emergency — while long queues breed frustration and conflict. Based on the results, optimized passenger routing using modern technology is the most promising strategy of those tested.

TO EXPAND ON WHEN PRESENTING

- Why overcrowding is a risk, not just a comfort issue
- What the simulation model represents and tests
- The strategies compared for managing crowds
- Why routing beat the other strategies tested
- Keywords: crowd management · risk assessment · simulation model

800

passengers per hour modeled for the sample terminal

